THREAT-INTELLIGENCE KILL-CHAIN CORRELATION ENGINE

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ABSTRACT

The Threat-Intelligence Kill-Chain Correlation Engine is a cybersecurity system designed to analyze and correlate security events from multiple sources to identify complete attack sequences instead of isolated alerts. Modern networks generate thousands of security logs, making it difficult for analysts to detect real threats efficiently. This project collects security events, applies advanced data structures and algorithms such as Graphs, Tries, KMP, and Rabin-Karp for efficient pattern matching and event correlation, and reconstructs the attacker's progression through the cyber kill chain. The system visualizes attack paths, prioritizes threats based on severity, and provides real-time insights through an interactive dashboard. By reducing alert fatigue and improving threat detection accuracy, the proposed solution helps security teams respond faster to cyber incidents. The project demonstrates the practical application of data structures and algorithms in cybersecurity while providing a scalable foundation for future integration with IoT-based security monitoring systems and AI-driven threat intelligence.

INTRODUCTION

- Modern networks generate thousands of security alerts.
- Individual alerts may not reveal the complete attack.
- Our system connects related security events.
- Reconstructs the attacker's progression.
- Helps security teams detect and respond faster.

SOFTWARE & HARDWARE REQUIREMENTS

Software:

- Operating System: Windows / Linux
- Programming Language: Python
- Database: MongoDB
- IDE: VS Code
- Web Dashboard: HTML, CSS, JavaScript

Hardware:

- Intel Core i5 or equivalent
- 8 GB RAM
- 256 GB Storage
- 64-bit System
- Internet Connection

LITERATURE SURVEY

- Existing systems generate and analyze security alerts.
- Threat-intelligence systems identify suspicious indicators.
- Graph-based systems represent relationships between events.
- Pattern-matching algorithms detect known patterns.
- Research Gap: Difficulty in correlating multiple alerts into a complete attack sequence.

Our approach:

Combine event correlation + data structures + Kill Chain analysis.

METHODOLOGY

Security Events

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Data Processing

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Pattern Matching
(KMP, Rabin-Karp, Trie)

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Event Correlation
(Graph)

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Kill-Chain Reconstruction

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Threat Prioritization

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Dashboard Visualization



Thank You

Stay Safe, Stay Smart